

Database connection pool exhausted under spike load

Issue Description

During peak traffic hours, the PostgreSQL backend database throws `OperationalError` too many connections and transaction deadlocks occur on high concurrency write workloads.

Steps to Reproduce

1. Execute 100 concurrent PATCH requests to user settings endpoint
2. Observe database lock contention

Expected vs Actual Behavior

Expected: Database handles locks sequentially without throwing `OperationalError`.

Actual: Database transaction deadlocks with `PGError 40P01`.

Environment Metadata

PostgreSQL 16, SQLAlchemy 2.0, FastAPI

Error Stack Trace / System Logs

```
sqlalchemy.exc.OperationalError: (psycopg2.errors.DeadlockDetected) ERROR: deadlock detected  
DETAIL: Process 1820 waits for ShareLock on transaction 94820; blocked by process 1821.
```